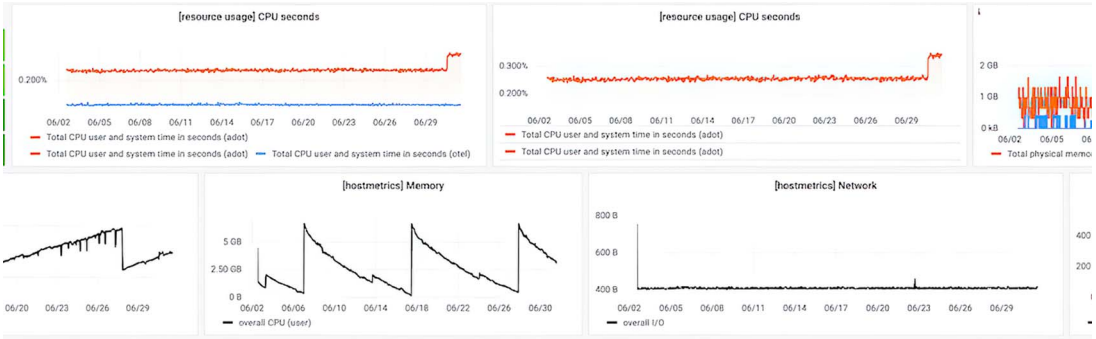


Figure 4.11 Grafana dashboard of OpenTelemetry collector resource usage

4.5.3 Agent nonfunctional requirements

In addition to the functional aspects, it's important to ask yourself the following:

- Is the agent open source, extensible, or backed by a single vendor or community?
- Who watches the watcher, or, put differently, what about the agent telemetry? In 2023 (and beyond), you should expect not only to have access to the agent logs but also get metrics (ideally, in Prometheus format) as well as traces and profiles from your agent.
- Is the agent capable of dealing with one or more signal types? A signal-specific agent means you need multiple agents for 360 coverage.

There is much more to nonfunctional requirements than we've discussed, but again, these questions form the basics, and you may choose to extend them with organizational requirements or preferences when selecting an agent.

Summary

- There are two types of agents: vendor-specific ones, like the Datadog agent, and signal-specific ones (usually open source), like Fluent Bit for logs.
- In this book, we consider vendor-specific and signal-specific agents as traditional agents, in contrast to the open source, all-encompassing OpenTelemetry collector.
- OpenTelemetry is a set of specifications, SDKs, a protocol (OTLP), and agents.
- OpenTelemetry allows you to collect, process, and ingest logs, metrics, traces, and (in the future) profiles in a vendor-neutral way.
- With OpenTelemetry, you get rich metadata both on where the signals came from (resource attributes) and about the telemetry signals themselves.
- Based on the semantic conventions, OpenTelemetry enables you to implement effective correlation in the backends.